

SUBJECT

B-52 Stratofortress-class airframe

source geometry: b52.stl

OBJECTIVE	Solve the drag on this airframe and paint the surface pressure field.
SOLVER & MODEL	OpenFOAM simpleFoam - k-omega SST, steady RANS
ISSUED (UTC)	2026-07-21T18:04:00Z

RESULT

SOLVER-BACKED

Drag coefficient

0.0471 ± 0.0018

95% confidence interval

Basis: steady RANS on a watertight airframe with no in-regime experimental anchor for this configuration; the reference area is the measured silhouette (601 m^2), not the published wing area

Lift coefficient	0.0192 0.0041
Mesh	193,880 cells max non-orthogonality 62, max skewness 3.1

UNCERTAINTY - ASME V&V 20 THREE-CHANNEL

CHANNEL	VALUE	STATE
Input	48.5 m ref. length	quantified
	reference length stated, not inferred; freestream at cruise	
Numerical	grid difference	not quantified
	single mesh; a two-mesh probe yields a grid difference, not a verified GCI (Eca & Hoekstra)	
Model form	fully-turbulent RANS	not quantified
	k-omega SST, no transition model; Aref is the measured silhouette, so Cd is not directly comparable to published book values	

PROVENANCE - EVIDENCE-BUNDLE SEAL (SHA-256)

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Mission geometry-study-b52 - Certificate C-2026-4658